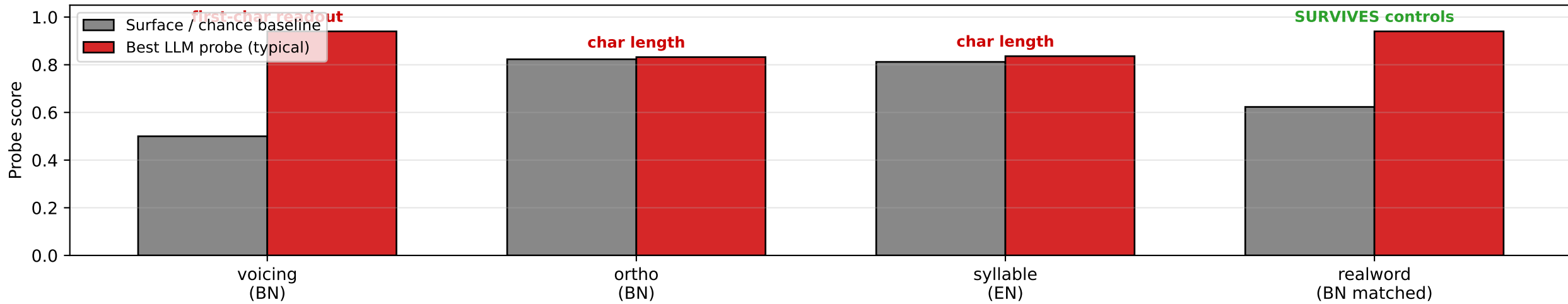


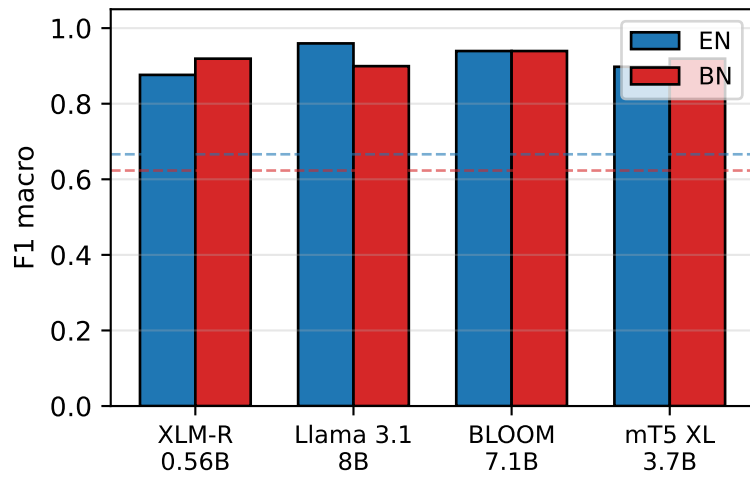
DysLex-Probe: Cross-Lingual Phonological Probing of Four Multilingual LLMs

Surface-Confound Sanity Checks → Length-Matched Recovery → Causal Mechanism

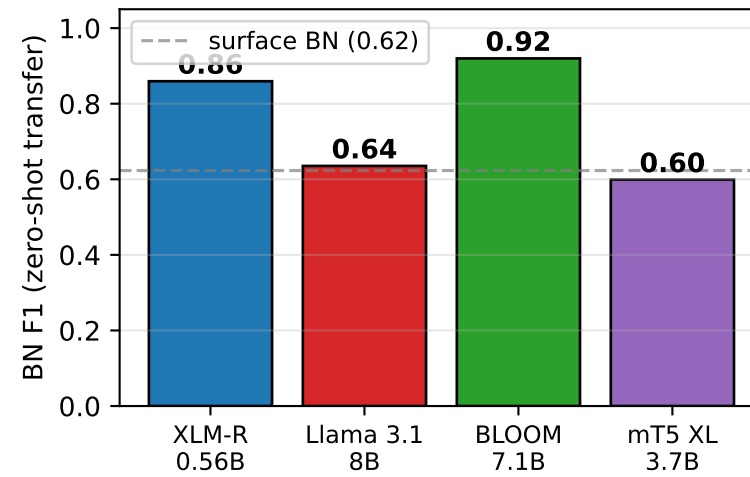
A. Three of four "phonological" probing tasks fail surface-feature sanity checks



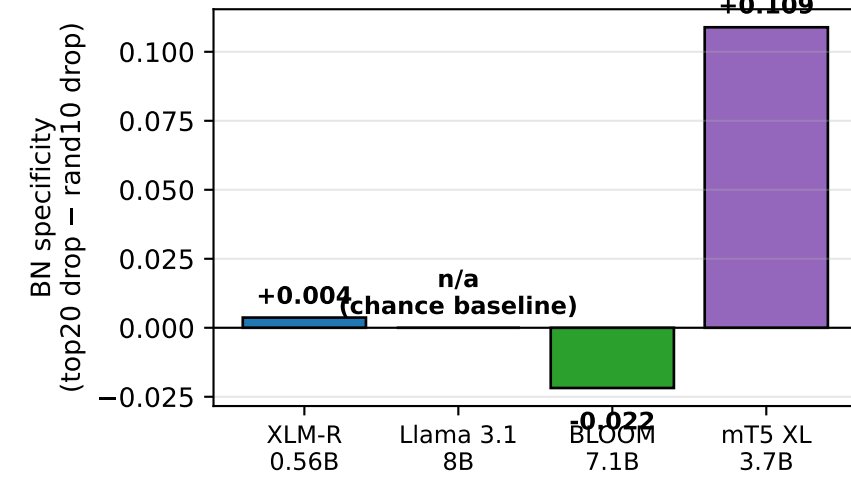
B. Within-language realword (length-matched)
All models exceed surface baseline



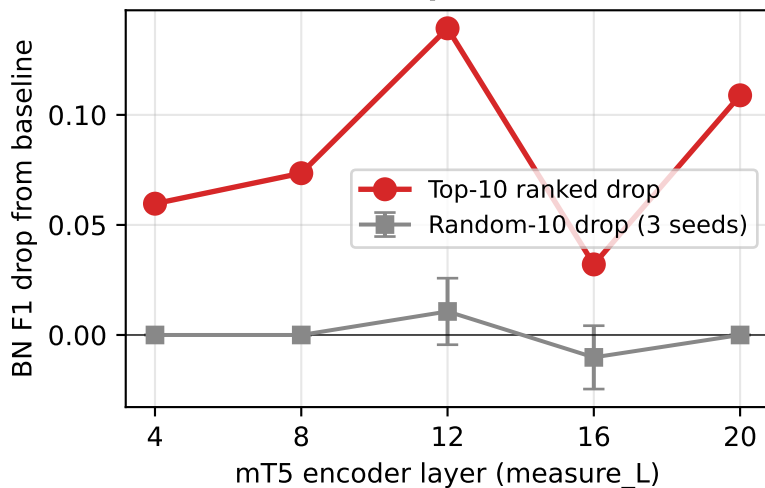
C. Zero-shot EN→BN transfer (length-matched)
BLOOM + XLM-R survive; Llama + mT5 collapse



D. Causal ablation: only mT5 localizes lexical info to specific units



E. mT5 localization persists across encoder depth



F. mT5 hybrid mechanism: two persistent "backbone" units (red, ≥3 layers) + layer-specialized circuits (gray)

